

SOC ANALYST PORTFOLIO

Hands-On Splunk SIEM Investigations
and Detection Engineering

5

Documented Investigations

2

Engineered Detection Rules

3

Tracked Detection Fires (Verified)

0

False Positives on Engineered Rules

7

MITRE ATT&CK Techniques Mapped

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1. Lab Architecture Overview

This portfolio documents work conducted in a self-built SOC home lab. The lab simulates a small enterprise environment with a centralized SIEM (Security Information and Event Management platform) collecting Windows endpoint telemetry through a forwarder agent. All investigations and engineered detections in this portfolio were executed in this environment.

Stack

Component	Details
Hypervisor	VMware Workstation 17 Player
SIEM Platform	Splunk Enterprise 9.1.2 on Ubuntu Server VM
Endpoint	Windows 10 (hostname: WIN-CLIENT-01) with Splunk Universal Forwarder
Endpoint Logging	Windows Security Event Log; Sysmon with SwiftOnSecurity config
Forwarder Service	SplunkForwarder running as LocalSystem
Networking	NAT topology between host, Splunk VM, and endpoint VM
Resource Allocation	Splunk VM: 4 GB RAM allocation; host: 16 GB total

Verified Telemetry Sources

- Windows Security Event Log forwarded from WIN-CLIENT-01 to Splunk indexer
- Sysmon (Microsoft Sysinternals, captures detailed process, network, and registry activity beyond native Windows logging)
- Sysmon EventCode 1 (Process Create) confirmed flowing into Splunk
- Sysmon EventCode 8 (CreateRemoteThread) confirmed flowing into Splunk

Design Notes

VMware Workstation 17 Player does not support host-only networking with the Player edition; NAT was the production-acceptable substitute. The SplunkForwarder service account was set to LocalSystem to resolve Access Denied / errorCode=5 errors when reading the Sysmon event channel. Resource allocation is constrained on a 16 GB host: only two VMs run concurrently in active investigations, with the indexer optionally offline during endpoint preparation phases (the Universal Forwarder queues events locally during indexer downtime).

2. Skills Demonstrated

SIEM and Log Analysis

- Splunk Search Processing Language (SPL) for triage, baseline checks, and detection authoring
- Splunk alert engineering with real-time and scheduled triggers, throttling, and Triggered Alerts tracking
- Log ingestion configuration: Splunk Universal Forwarder, Windows Event Log, Sysmon channel

Investigation Disciplines

- Baseline behavior analysis: confirming whether observed activity matches a user, host, or service account's historical pattern
- Pivoting across event sources to construct event timelines
- False positive identification and tuning recommendations
- Triage decision frameworks: false positive closure, escalation, host containment

Detection Engineering

- Authoring SPL detections mapped to MITRE ATT&CK techniques
- Detection validation through controlled attack simulations (PowerShell, registry writes)
- Documenting prerequisites (audit policy, SACL configuration) that determine whether a detection can fire
- Tracking verified fires with timestamps and false positive rates

MITRE ATT&CK Coverage Across This Portfolio

Technique ID	Technique Name	Where Applied
T1110	Brute Force	Investigation 3, Engineered Detection 1
T1078.002	Valid Accounts: Domain Accounts	Investigation 4 (lateral movement)
T1021.002	Remote Services: SMB / Admin Shares	Investigation 4
T1059.003	Command and Scripting Interpreter: Windows Command Shell	Investigation 4
T1098	Account Manipulation	Investigation 4
T1033	System Owner / User Discovery	Investigation 4
T1547.001	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder	Investigation 5, Engineered Detection 2

3. Investigation 1: EventCode 4702 Scheduled Task False Positive

Case ID	INC-2024-0012
EventCode	4702: Scheduled task updated
Severity at Alert	Medium
Source Host	WIN-CLIENT-01
Triage Outcome	False Positive (closed with tuning recommendation)
Analyst	Eyimofe Samuel Olaiya

Why the Alert Fired

EventCode 4702 is generated when a scheduled task is updated in the Windows Task Scheduler. Scheduled task modifications are a known persistence technique (MITRE T1053.005), so any 4702 entering the queue warrants triage to confirm whether the modification was authorized.

Investigation Steps

- Step 1: Pulled the 4702 event details and identified the calling user context, the task name, and the modification timestamp
- Step 2: Cross-referenced the calling account against the known service account list for the endpoint
- Step 3: Pulled prior 4702 events for the same task name across the historical index to establish whether the modification was a recurring scheduled action
- Step 4: Confirmed the modification time matched the expected cron window for the patch management tool that owns the task

Findings

- The modification originated from a known IT patch management tool running on its documented schedule
- The user context matched the service account the tool always runs as
- The modification timestamp matched the documented run window
- The task name was on the documented allow-list for this endpoint

Disposition and Tuning Recommendation

Closed as false positive. Recommended tuning: suppress 4702 events where the calling subject account matches the patch management service account and the modified task name appears on the allow-list. This reduces predictable noise without losing visibility on out-of-band 4702 events from other sources.

Analyst Note. False positives are not failures. A correctly tuned 4702 detection still fires on this exact pattern by design until the tuning rule is applied. Documenting the cause is what enables the next analyst to recognize the pattern in seconds, or to write the suppression rule that eliminates the noise entirely.

4. Investigation 2: EventCode 5379 Credential Manager False Positive

Case ID	INC-2024-0019
EventCode	5379: Credential Manager credentials were read
Severity at Alert	Medium
Source Host	WIN-CLIENT-01
Triage Outcome	False Positive (closed with tuning recommendation)
Analyst	Eyimofe Samuel Olaiya

Why the Alert Fired

EventCode 5379 logs when stored credentials in the Windows Credential Manager (the local secure store that holds saved passwords for sites, applications, and network resources) are read. Credential Manager reads can indicate credential theft attempts (MITRE T1555.004), so any 5379 entering the queue is worth a fast pivot.

Investigation Steps

- Step 1: Pulled the 5379 event details, noted the calling process and the user session in which the read occurred
- Step 2: Established the user's normal morning login pattern: time of day, machine, session origin
- Step 3: Identified the calling process as msedge.exe (Microsoft Edge browser) reading a saved credential to auto-fill the user's email login form
- Step 4: Confirmed the read occurred in the user's normal interactive session, on their normal device, with no anomalous parent process or network indicator

Findings

- The read was initiated by Microsoft Edge auto-filling a saved password at user login
- Session origin, device, and timing all matched the user's documented baseline
- No anomalous parent process, no unusual network connection, no secondary credential access events in the same window

Disposition and Tuning Recommendation

Closed as false positive. Recommended tuning: filter 5379 events where the calling process matches a known browser executable (msedge.exe, chrome.exe, firefox.exe) on user-owned endpoints. This eliminates predictable auto-fill noise while preserving 5379 alerts from non-browser processes, which retain investigative value.

5. Investigation 3: EventCode 4625 Brute Force Attack

Case ID	INC-2024-0031
EventCode	4625: Failed Logon
Severity at Alert	High
Source Host	WIN-CLIENT-01
Events Detected	21 EventCode 4625 events within a 1-minute window
Simulation Method	PowerShell script: 20 sequential failed logon attempts against account 'fakeuser'
Triage Outcome	True Positive: Detection Rule Fired
MITRE Technique	T1110 (Brute Force)
Analyst	Eyimofe Samuel Olaiya

Scenario

A brute force attack is one in which an adversary repeatedly attempts to authenticate using different passwords (or the same password against many accounts in a password-spray variant). This investigation simulated a credential stuffing pattern against a local Windows account and validated end-to-end detection in Splunk: log generation, forwarding, ingestion, query, and alert trigger.

Attack Simulation

A PowerShell script was executed on WIN-CLIENT-01 to generate 20 consecutive failed logon attempts against the non-existent local account 'fakeuser'. The script used a tight loop to produce rapid EventCode 4625 entries, mimicking the behavior of a brute force tool. Twenty-one 4625 events were observed in a single 1-minute window (the extra event was an additional 4625 generated by a separate secondary logon side effect, documented in Engineered Detection 1).

Detection Steps

- Queried all EventCode 4625 events for the simulation window and confirmed 21 events appeared in Splunk within the detection window
- Used SPL bucket + stats to count failed logons per account per 1-minute window, identifying the spike on 'fakeuser'
- Confirmed LogonType, IpAddress, and Account_Name fields populated correctly in the indexed events for analyst review

SPL Queries Used

```
host="WIN-CLIENT-01" EventCode=4625 | table _time, Account_Name, IPAddress, LogonType, Message | sort -_time
```

```
host="WIN-CLIENT-01" EventCode=4625 | bucket _time span=1m | stats count by _time, Account_Name | where count > 5
```

Detection Rule Built

A Splunk Real-Time Alert was built to operationalize this detection. Full rule configuration is documented in Engineered Detection 1.

Alert Name	Brute Force Detection: Multiple Failed Logons
Trigger Type	Real-time
Threshold	Greater than 5 failed logons against a single account in a 1-minute bucket
Throttle	Enabled (suppresses duplicate notifications)
MITRE	T1110

Triage Decision and Reasoning

Confirmed True Positive. The activity pattern (20+ sequential failed authentications against a single account from a single host inside one minute) has no legitimate explanation in normal user behavior. Even users repeatedly mistyping a password almost never generate this volume of failures inside a 60-second window.

Indicators of Compromise (IOCs)

Type	Value	Context
Account Name	fakeuser	Target of the brute force attempts
Source Host	WIN-CLIENT-01	Originating endpoint for the failed authentications
EventCode Spike	21 x 4625 in 1 minute	Volume characteristic of automated tool
Logon Type	2 (Interactive) for primary attempts	Console-style authentication failures

Lessons Learned

The Windows Secondary Logon service generates parallel 4625 events against both the target account and the launching session account under specific authentication conditions. This explains why a 20-attempt PowerShell loop produced 21 indexed 4625 events. A detection author who is not aware of this can be confused by event count mismatches between the attack script and the SIEM. This is documented in Engineered Detection 1 because it affects threshold tuning.

6. Investigation 4: Lateral Movement via Compromised Service Account

Case ID	INC-2024-0047
Alert Title	Lateral Movement: Admin Share Access and Credential Tampering
Severity	HIGH
Host	WKSTN-031
Account Involved	backup_svc (service account)
Source IP	10.10.50.14 (internal, unassigned asset)
Detection Time	23:51:44
Triage Decision	CONTAIN HOST
Status	Escalated to Tier 2 / Incident Response
Analyst	Eyimofe Samuel Olaiya

Alert Summary

At 23:51:44 the SIEM triggered a HIGH severity alert for suspicious activity on WKSTN-031. The backup_svc service account logged in over the network from an unrecognized internal IP (10.10.50.14), accessed the administrative C\$ share, then spawned cmd.exe from services.exe and executed a command attempting to modify the domain administrator password. This combination of behaviors - off-schedule access, admin share mount, child process from a service, and domain credential modification - was assessed as an active lateral movement incident.

Raw Event Timeline

Time	EventCode	Description	Key Details
23:51:44	4624	Successful Network Logon	backup_svc from 10.10.50.14, Logon Type 3
23:51:51	5140	Network Share Accessed	\\WKSTN-031\C\$: ReadData + WriteData
23:52:01	4688	Process Created	cmd.exe spawned by services.exe; net user administrator P@ss2024! /domain
23:52:14	4688	Process Created	whoami.exe spawned by cmd.exe

Investigation Steps and Findings

Step 1. Baseline Behavior Check

```
index=winsec host=WKSTN-031 user=backup_svc | stats count by EventCode, CommandLine | sort -count
```

- backup_svc has no historical record of network logons to WKSTN-031
- The account's normal activity is limited to FILESVR-01 during the 01:00 to 03:00 backup window
- No prior instances of cmd.exe or whoami.exe associated with this account
- Assessment: anomalous behavior with no legitimate baseline match

Step 2. Source IP Threat Intelligence Check

- IP 10.10.50.14 is within the internal 10.10.50.0/24 subnet
- IP is not assigned to any known asset in the CMDB (Configuration Management Database)
- Not the registered backup server IP
- No prior network activity recorded from this host before 23:51
- Assessment: internal threat - unknown or unauthorized asset acting as source of this logon

Step 3. Service Account Audit

- Scheduled to run only between 01:00 and 03:00 - tonight's activity started at 23:51, outside its window
- Account permissions scoped to read access on FILESVR-01 only - accessing WKSTN-031 admin share is outside authorized scope
- Account type is a service account - it should never interactively spawn cmd.exe or run net user commands
- Last legitimate recorded activity: 3 nights ago, within normal backup window
- Assessment: account credentials are likely compromised and being abused by a threat actor

Step 4. Network Trace and Lateral Activity Scope

- WKSTN-031 made outbound SMB (port 445) connections to DC-01 (Domain Controller) and FILESVR-01
- Timing: 23:52 to 23:59, approximately 7 minutes of active connections post-logon
- 34 KB outbound to 10.10.50.14 - possible data staging or C2 callback
- The DC-01 connection is critical - this suggests the attacker attempted to propagate the password change to Active Directory

MITRE ATT&CK Mapping

Technique ID	Name	Evidence
T1078.002	Valid Accounts: Domain Accounts	backup_svc used outside normal context
T1021.002	Remote Services: SMB / Admin Shares	\\WKSTN-031\C\$ mounted with write access
T1059.003	Windows Command Shell	cmd.exe spawned from services.exe
T1098	Account Manipulation	net user administrator P@ss2024! /domain
T1033	System Owner / User Discovery	whoami.exe executed post-access

Triage Decision and Reasoning

Decision: CONTAIN HOST (WKSTN-031), escalate to Tier 2 / Incident Response.

Why not False Positive?

There is no business justification for backup_svc to (a) log in at 23:51 outside its scheduled window, (b) access an admin share on a workstation it has no authorized scope over, (c) spawn an interactive command shell from a service process, and (d) attempt to modify the domain administrator password. All four behaviors would need to be explained simultaneously by a legitimate action. No change tickets or maintenance windows were open for this activity.

Why CONTAIN rather than just ESCALATE?

The command 'net user administrator P@ss2024! /domain' represents an irreversible action if it successfully replicates to Active Directory. Unlike a brute force attempt (where waiting for Tier 2 carries acceptable risk), allowing this host to remain active during incident review risks:

- Domain administrator credentials being changed, locking out the security team
- The attacker pivoting to DC-01 (network trace confirms the connection was already initiated)
- Additional hosts being compromised from WKSTN-031 as a staging point

The risk of inaction outweighs the disruption of containment. Containment is reversible in 30 seconds; an executed domain administrator password change is not.

Escalation Note to Tier 2

Escalation summary. WKSTN-031 has been isolated from the network. backup_svc credentials should be rotated immediately and audited across all systems. 10.10.50.14 is an unknown internal asset; physical location and owner should be identified. DC-01 event logs should be reviewed for any successful password replication from this incident window (23:51 to 23:59). Recommend full forensic image of WKSTN-031 before restoration.

Key Indicators of Compromise (IOCs)

Type	Value	Context
Internal IP	10.10.50.14	Unknown asset, source of suspicious logon
Account	backup_svc	Compromised service account
Process Chain	cmd.exe (parent: services.exe)	Abnormal parent-child relationship
Command	net user administrator P@ss2024! /domain	Domain credential tampering
Share	\\WKSTN-031\C\$	Admin share accessed outside authorized scope

Lessons Learned

A service account accessing a network share is not inherently suspicious; that is what backup accounts do. What makes this incident stand out is the combination of anomalies, not any single one: wrong time + wrong host + wrong process + wrong command = confirmed threat.

This is the analyst skill that matters most at Tier 1: recognizing that individual events can be explained away, but correlated anomalies cannot. The more boxes that do not fit the baseline, the higher the confidence that something is wrong.

7. Investigation 5: Persistence via Registry Run Key Modification

Case ID	INC-2024-0048
Alert Title	Persistence: Registry Run Key Write
Severity	HIGH
Status	Resolved (Lab Simulation)
Host	WIN-CLIENT-01
Account Involved	socadmin
Detection Time	2026-04-16 14:57:44 UTC
MITRE Technique	T1547.001 (Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder)
Analyst	Eyimofe Samuel Olaiya

Executive Summary

A registry modification event was detected on endpoint WIN-CLIENT-01 targeting the Windows Run key, a common persistence mechanism used by attackers to survive system reboots. The event was captured via Windows Security auditing, forwarded to Splunk via Universal Forwarder, and detected using a custom SPL query. The entry was identified as a simulated malicious payload planted under a deceptive process name designed to blend in with legitimate Windows activity.

Attack Overview

The Windows Run key is a built-in Windows feature that automatically executes any program listed under it every time a user logs in. Attackers abuse this to maintain persistent access: even if the machine is rebooted or the user logs out and back in, the malware continues to execute silently.

This technique requires no special privileges on a standard user account, making it accessible to even unsophisticated attackers. It is one of the oldest and most widely used persistence mechanisms in Windows environments.

Registry path targeted

```
HKKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run
```


Detection Source

Field	Value
SIEM	Splunk 9.1.2
Log Source	Windows Security Event Log
Event Code	4657: A registry value was modified
Index	main
Host	WIN-CLIENT-01
Detection Time	2026-04-16 14:57:44 UTC

Splunk Detection Query

```
index=main source="WinEventLog:Security" EventCode=4657 "CurrentVersion\Run" | table _time, ComputerName, Account_Name, Object_Value_Name, New_Value, Process_Name
```

Event Details

Field	Value
Time	2026-04-16 14:57:44.653
Computer Name	WIN-CLIENT-01
Account Name	socadmin
Object Name	\REGISTRY\USER\<SID>\Software\Microsoft\Windows\CurrentVersion\Run
Object Value Name	WindowsUpdateService
Operation Type	New registry value created
Process Name	reg.exe (interactive write)

Investigation Steps

- Step 1: Confirmed Object Access audit policy was enabled and SACL (System Access Control List, the per-object audit permission) was applied to the Run key, allowing 4657 events to fire on registry writes
- Step 2: Pulled the 4657 event details, identified the new value name 'WindowsUpdateService' written to HKCU\...\CurrentVersion\Run
- Step 3: Recognized the value name as a deceptive label - 'WindowsUpdateService' is not a standard Microsoft autorun entry name

- Step 4: Confirmed no legitimate software install or Windows Update activity was occurring on the host in this window
- Step 5: Validated detection logic with a controlled HKCU Run key write, producing the expected 4657 event

MITRE ATT&CK Mapping

Technique ID	Name	Evidence
T1547.001	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder	New value written to HKCU CurrentVersion\Run with deceptive name

Triage Decision

Confirmed True Positive (simulated). In a production environment this disposition would route to a containment workflow: kill any running instances of the autorun target, remove the registry value, image the host for analysis, and rotate credentials for socadmin.

Lessons Learned

EventCode 4657 does not fire by default. Two prerequisites must be in place for the detection to work: Object Access auditing must be enabled in the audit policy, and a SACL must be applied to each registry key you wish to monitor. Without both, a Run key write is invisible to the Windows Security log. This prerequisite is the most common reason registry-based detections fail to trigger when copied from a runbook into a new environment, and it must be documented alongside the SPL query.

8. Engineered Detection 1: Brute Force, Multiple Failed Logons

Rule Name	Brute Force Detection: Multiple Failed Logons
Trigger Type	Real-time alert
Data Source	Windows Security Event Log via Splunk Universal Forwarder
EventCode Monitored	4625: Failed Logon
MITRE Technique	T1110 (Brute Force)
Verified Date(s)	April 10, 2026 (initial simulation); April 23, 2026 (re-verification)
Tracked Fires (April 23)	2 (at 13:27:26 EDT and 13:30:12 EDT)
False Positives	0
Throttle	Enabled

Detection Logic

Fires when a single account on a single host exceeds 5 failed logons (EventCode 4625) within a 1-minute time bucket. The thresholds are tuned to capture automated brute force and credential stuffing patterns while ignoring normal human typo behavior, which almost never produces 5 failures inside 60 seconds on the same account.

SPL Query

```
index=winsec EventCode=4625 host=WIN-CLIENT-01 | bucket _time span=1m | stats count by _time, Account_Name | where count > 5
```

Splunk Alert Configuration

Setting	Value
Trigger condition	Number of results greater than 0
Trigger schedule	Real-time (continuous)
Throttle window	Enabled on Account_Name + host
Triggered Alerts tracking	Enabled
Severity	High
Action	Log alert; notification on Triggered Alerts dashboard

Verification Evidence

Two re-verification fires on April 23, 2026 from a controlled PowerShell-driven authentication burst:

- Fire 1: 2026-04-23 13:27:26 EDT - threshold exceeded against target account
- Fire 2: 2026-04-23 13:30:12 EDT - threshold exceeded against target account in a second test wave

Both fires matched the expected account and host. Zero false positives across both fires. An earlier full simulation on April 10, 2026 produced 21 EventCode 4625 events in a single 1-minute window and triggered the same logic historically when the rule was re-run against the indexed archive.

Why a 1-Minute Bucket and a Threshold of 5

- Normal human typo behavior almost never reaches 5 failures inside one minute on the same account; 5 is conservative without being noisy
- Time windows shorter than one minute risk splitting a single attack across two buckets and missing the alert
- The tradeoff is alert latency: up to a one-minute delay on the trigger is acceptable in exchange for not splitting attacks
- Production tuning should use real data rather than lab data to refine thresholds for the local environment

Known False Positive Scenarios

Service accounts whose configuration breaks at the same moment can generate rapid auth failures that would fire this rule. Two mitigations are available:

- Exclude known service accounts from the rule entirely
- Split the rule so service accounts have a higher threshold (10+) and human accounts have the standard threshold (5)

The current rule does not implement the split. In a production rollout, the split should be implemented before the rule is enabled.

What This Detection Does Not Catch

- Password spraying: many different usernames each receiving few failures - same source attempting many accounts looks different statistically and requires a separate detection
- Slow brute force: attempts spaced beyond a minute to evade volume-based thresholds
- Successful brute force endpoint: this rule fires on failed authentications only; pairing with a 4624 watch for the same account post-spike would close the loop

Lessons Learned

During testing, the Windows Secondary Logon service generated parallel 4625 events against both the target account and the launching session account under specific authentication conditions, effectively doubling the event count for some attack patterns. A 20-attempt PowerShell loop produced 21 indexed events. This is a genuine Windows authentication behavior, not a bug in the detection, and it does not change the rule logic but it does explain the small variance between attack volume and indexed event volume in the verification runs.

9. Engineered Detection 2: Persistence, Registry Run Key Modified

Rule Name	Persistence Detection: Registry Run Key Modified
Trigger Type	Scheduled alert (hourly)
Data Source	Windows Security Event Log via Splunk Universal Forwarder
EventCode Monitored	4657: Registry value modified
MITRE Technique	T1547.001
Verified Date(s)	April 16, 2026 (initial simulation); April 23, 2026 (re-verification)
Tracked Fires (April 23)	1 (at 15:09:50 EDT)
False Positives	0
Prerequisite	Object Access auditing enabled + SACL applied to monitored keys

Detection Logic

Fires on EventCode 4657 events where the Object Name contains the substring 'CurrentVersion\Run'. This single substring covers all four registry paths attackers most commonly use for autorun-based persistence:

- HKLM\Software\Microsoft\Windows\CurrentVersion\Run
- HKLM\Software\Microsoft\Windows\CurrentVersion\RunOnce
- HKCU\Software\Microsoft\Windows\CurrentVersion\Run
- HKCU\Software\Microsoft\Windows\CurrentVersion\RunOnce

SPL Query

```
index=winsec EventCode=4657 Object_Name="*CurrentVersion\Run*"
```

Splunk Alert Configuration

Setting	Value
Trigger condition	Number of results greater than 0
Trigger schedule	Scheduled, every 1 hour
Triggered Alerts tracking	Enabled
Severity	High
Action	Log alert; notification on Triggered Alerts dashboard

Why Scheduled Hourly, Not Real-time

- Registry writes to Run keys are a low-frequency event in normal usage; an hour of latency is acceptable
- Real-time detections cost more on the indexer and produce more triage interrupts
- For persistence specifically, the attacker has already established the foothold by the time the alert is seen; most value is in post-incident triage and confirming the IoC is preserved
- If the detection does not need real-time to be useful, it should not be real-time

Prerequisites for the Detection to Fire

Critical prerequisite. EventCode 4657 does not log by default in Windows. Two configuration steps are required: enable the 'Audit Registry' subcategory under the Object Access audit policy, and apply a SACL (System Access Control List) to each registry key that should be monitored. Without both, no 4657 events will be generated and the detection will silently never fire.

These prerequisites were configured on WIN-CLIENT-01 before this detection was deployed. They must be replicated on any other host where this rule is intended to provide coverage.

Verification Evidence

Re-verification on April 23, 2026 against a controlled HKCU Run key write:

- Fire: 2026-04-23 15:09:50 EDT - 4657 event matched the substring pattern, alert triggered on the next scheduled run

An earlier simulation on April 16, 2026 (the same incident documented as INC-2024-0048) produced 5 matching 4657 events from a sequence of registry writes during the attack scenario. Zero false positives observed across both verification runs.

What This Detection Does Not Catch

- Persistence via Startup folder files - would require file-system auditing or Sysmon EventCode 11
- Persistence via Scheduled Tasks - monitored by EventCode 4698 (separate detection)

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- Persistence via Windows services - monitored by EventCode 7045 in the System log (separate detection)
 - Persistence via WMI event subscriptions - not visible in Windows Security log; requires WMI-Activity operational log or Sysmon EventCode 19/20/21

The Run key detection is one technique-specific control. A complete persistence detection strategy requires multiple rules covering the techniques above in addition to this one.

Improvement Roadmap

- Migrate to Sysmon EventCode 13 (RegistryEvent value set) as the data source - removes the SACL configuration requirement, simpler to deploy across many endpoints
- Enrich the detection with parent process information to surface which process wrote the Run key (filter on uncommon writers)
- Pair with an autorun executable hash check to identify which binary the Run entry points to

10. Summary of Results and MITRE Coverage

Investigation Outcomes

Case	Type	Outcome	Closure Time
INC-2024-0012 (EID 4702)	Triage	False Positive, tuning recommended	Closed
INC-2024-0019 (EID 5379)	Triage	False Positive, tuning recommended	Closed
INC-2024-0031 (EID 4625)	Detection validation	True Positive, rule fired	Closed
INC-2024-0047 (Lateral Movement)	Active incident	Host contained, escalated to Tier 2	~18 minutes
INC-2024-0048 (Persistence)	Detection validation	True Positive (lab simulation)	Closed

Detection Rule Verification Snapshot

Rule	Trigger	Verified Fires	FP Count	MITRE
Brute Force: Multiple Failed Logons	Real-time	2 (April 23, 2026)	0	T1110
Persistence: Registry Run Key Modified	Hourly	1 (April 23, 2026)	0	T1547.001

Aggregate Metrics

Metric	Value
Total documented investigations	5
Total engineered detection rules	2
Total tracked detection fires (verified)	3
Total false positives on engineered rules	0
MITRE ATT&CK techniques mapped	7 across portfolio
Mean triage-to-decision time (active incident)	Approx. 18 minutes (INC-2024-0047)

Analyst Methodology

Every investigation in this portfolio follows the same five-step process: validate the alert against data source integrity; establish baseline context for the user, host, and behavior; pivot across correlated events to construct the timeline; decide on disposition (false positive, escalation, or containment); and document the case with IOCs, MITRE mapping, query trail, and escalation notes. This is the discipline that transfers from a lab environment to an enterprise SOC queue.

11. Contact

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Areas of Focus

- Splunk SIEM operations, SPL authoring, alert engineering
- Windows Security event log analysis (authentication, process, registry, share access)
- Sysmon-based endpoint telemetry
- Investigation triage and incident escalation
- MITRE ATT&CK technique mapping and detection coverage analysis

Currently Building Toward

- Active Directory domain detection work (Kerberoasting, privilege escalation, Group Policy abuse)
- Microsoft Sentinel and KQL (Kusto Query Language) for cloud SIEM coverage
- CompTIA Security+ (SY0-701) certification

This portfolio represents independent, hands-on lab work. All investigations, detection rules, verification fires, and false-positive counts documented here are real outputs from the Splunk environment described in Section 1.